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Off-axis laser beam measurement for laser powder bed fusion

Abstract

Disclosed herein is a laser beam measurement system that includes a beam measurement device, a positioning apparatus, and a control system. The beam measurement device can include an imager and a plurality of optical elements configured to form an optical path to the imager within the beam measurement device. The positioning apparatus is configured to adjust a position of the beam measurement device relative to a build plate of a laser powder bed fusion system. The control system is configured to determine an intended measurement position to observe a laser beam profile of the laser powder bed fusion system, determine a target position of a beam measurement device to align with the intended measurement position, and command an adjustment of one or more aspects of the beam measurement device to achieve alignment of the target with the intended measurement position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/335,906 filed Apr. 28, 2022, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

(1) Exemplary embodiments of the present disclosure pertain to the art of metal additive manufacturing, and more particularly to off-axis laser beam measurement for laser powder bed fusion.

(2) Traditional additive manufacturing systems include, for example, Additive Layer Manufacturing (ALM) devices, such as Direct Metal Laser Sintering (DMLS), Selective Laser Melting (SLM), Laser Beam Melting (LBM) and Electron Beam Melting (EBM) that provide for the fabrication of complex metal, alloy, polymer, ceramic and composite structures by the freeform construction of the workpiece, layer-by-layer. The principle behind additive manufacturing processes involves the selective melting of atomized precursor powder beds by a directed energy source, producing the lithographic build-up of the workpiece. The melting of the powder occurs in a small localized region of the energy beam, producing small volumes of melting, called melt pools, followed by rapid solidification, allowing for very precise control of the solidification process in the layer-by-layer fabrication of the workpiece. An example of a particular type of system is a Laser Powder Bed Fusion (LPBF) additive system, where the energy beam is a laser.

(3) The laser in a LPBF system is typically focused by a lens, then deflected by a mirror so that the energy beam selectively impinges on a powder bed. In operation, the powder is melted at the laser focus site on the build surface or substrate. Precursor powder is either gravitationally fed from cassettes or loaded by a piston so that it can be raked onto a build table. The excess powder is raked off and collected for re-application. Since the laser is typically fixed, when additive manufacturing techniques are used in a repair or restoration operation, a single part may be processed at a time. Batch processing of multiple parts typically involves machining the parts down to a lowest common part dimension shared by the batch of parts and then adding to all of the parts, layer-by-layer, collectively to rebuild and restore the parts as a group.

BRIEF DESCRIPTION

(4) Disclosed is a laser beam measurement system that includes a beam measurement device, a positioning apparatus, and a control system. The beam measurement device can include an imager and a plurality of optical elements configured to form an optical path to the imager within the beam measurement device. The positioning apparatus is configured to adjust a position of the beam measurement device relative to a build plate of a laser powder bed fusion system. The control system is configured to determine an intended measurement position to observe a laser beam profile of the laser powder bed fusion system, determine a target position of a beam measurement device to align with the intended measurement position, and command an adjustment of one or more aspects of the beam measurement device to achieve alignment of the target with the intended measurement position.

(5) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the adjustment of one or more aspects of the beam measurement device can include the positioning apparatus changing an angle of the beam measurement device relative to the build plate.

(6) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, an optical path length within the beam measurement device can be adjustable to match an intended path length of a laser beam to the intended measurement position.

(7) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the adjustment of one or more aspects of the beam measurement device can include moving one or more of the optical elements.

(8) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the adjustment can maintain a path length of the optical path within the beam measurement device to accommodate a change in the target position.

(9) In addition to one or more of the features described above, or as an alternative to any of the

foregoing embodiments, the adjustment can maintain a path length of the optical path within the beam measurement device to accommodate a change in the target position.

(10) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the optical elements can include one or more semi-transparent mirrors and one or more beam dumps.

(11) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, a path length of the optical path within the beam measurement device can be greater than a height of a part of the beam measurement device from a build plane.

(12) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the control system can be configured to output the command to a user interface.

(13) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the control system can be configured to analyze the laser beam profile at the target position as observed by the imager, compare the laser beam profile to an expected beam profile, and trigger an alert based on determining that the laser beam profile fails to match the expected beam profile within a comparison threshold.

(14) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the beam measurement device can be integrated into a recoater of the laser powder bed fusion system.

(15) Also disclosed is a method for laser beam measurement in a laser powder bed fusion system. The method includes determining an intended measurement position to observe a laser beam profile of the laser powder bed fusion system by a beam measurement device, determining a target position of the beam measurement device to align with the intended measurement position, and commanding an adjustment of one or more aspects of the beam measurement device to achieve alignment of the target with the intended measurement position. A positioning apparatus is configured to adjust a position of the beam measurement device relative to a build plate of the laser powder bed fusion system.

(16) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the method can include outputting the command to a user interface.

(17) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the method can include operating the beam measurement device to observe the laser beam profile after adjusting one or more aspects of the beam measurement device, analyzing the laser beam profile at the target position as observed by the imager, comparing the laser beam profile to an expected beam profile, and triggering an alert based on determining that the laser beam profile fails to match the expected beam profile within a comparison threshold.

(18) A technical effect of the apparatus, systems and methods is achieved by performing off-axis laser beam measurement for laser powder bed fusion as described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

(2) FIG. 1 is a schematic view of a laser powder bed fusion system according to an embodiment;

(3) FIG. 2 is a schematic view of a measurement system according to an embodiment;

(4) FIG. 3 is a schematic view of a build plate of a laser powder bed fusion system that includes multiple build locations according to an embodiment;

(5) FIG. 4 is a schematic view of an integrated beam measurement device incorporated into a recoater of a laser powder bed fusion system according to an embodiment;

(6) FIG. 5 is a schematic view of an optical path within a beam measurement device according to an embodiment; and

(7) FIG. 6 is a process flow according to an embodiment.

DETAILED DESCRIPTION

(8) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

(9) Additive manufacturing has resulted in metal parts with decreased weight, added functionality and new design possibilities. Deposition method parameters may be selected to yield a desired average grain size. In some embodiments the average grain size is varied by location. In some embodiments the average grain size is consistent throughout. The thickness of the deposited layer can vary depending on the deposition method and desired average grain size.

(10) When directed energy deposition is used, laser power, toolpath speed, and layer thickness can be selected to yield the desired average grain size and layer thickness. In general, lower laser power and slower toolpath speed results in a smaller average grain size and higher laser power and faster toolpath speed results in a larger average grain size.

(11) Exemplary metals used in additive manufacturing can include nickel, cobalt, copper, titanium, aluminum, iron, alloys including one or more of the foregoing metals, and combinations thereof. Exemplary alloys can include Inconel 718, Inconel 625, stainless steel, Waspaloy and IN100. Metal powder particle size and particle size distribution in powder can be determined by one of skill in the art.

(12) Additive manufacturing can be used to build new parts or to repair damaged parts. The powder-bed fusion method allows for batch processing of multiple parts within a build volume. Selective laser sintering/melting in a powder bed can be used to perform weld build-up, for example. If the laser beam profile is not as intended, the weld pool (melt pool) will not be the correct shape, potentially inducing defects in the part. Commercial beam measurement devices are typically intended to be operated at a single location on a print platform of a LPBF machine, which is not sufficient where parts are printed across the print platform. Different locations on the print platform may have different beam profiles and temporal responses to beam irradiance based on several factors, such as, aberrations and astigmatism in laser optics, cleanliness of laser optics (especially build chamber laser entry window), inadequate calibration of focusing optics and/or positioning mirrors, and the like. Time varying beam profile can change due to thermal lensing, which can result in a change in focal distance of optics due to heating. For multi-laser systems, each beam may have a different profile at a given location on the print platform, leading to inconsistent print results depending on which laser is used. The lasers may also lose alignment relative to each other, potentially leading to defects in parts and/or increased surface roughness. Embodiments of the disclosure can include hardware and software that support positioning of a beam measurement device and associated optic elements to perform laser beam profile measurement at a location in space that accommodates factors such as machine design and optical variations to adjust one or more aspects of the beam measurement device to achieve alignment of a target with an intended measurement position. For example, performing laser beam profile measurement at a non-optimized depth position may not align with the beam focal point at the melt pool.

(13) FIG. 1 shows a schematic view of a laser powder bed fusion system **100** according to an embodiment as an example of an additive manufacturing system. The laser powder bed fusion system **100** includes an energy source **102**, a lens system **103**, one or more mirrors **105**, positioning apparatus **151** and a build tank **106**. The build tank **106** includes a base plate **108**. The base plate **108** can be adjustable to reposition one or more parts **112** as the parts **112** are built or repaired in a powder bed **114** of the build tank **106**. Alternatively, the build plate **108** and one or more parts **112** can remain in a fixed position. The energy source **102** can emit a laser **126** (also referred to as laser beam **126**) to form a new layer **128** of material fused on an uppermost level of the parts **112**. One or

more of the mirrors **105** can be adjusted to change the target location of the laser **126**. Since the build location of the one or more parts **112** within the build tank **106** can change, beam measurement systems that inspect the laser **126** before and/or after use at a fixed position, such as the center of the build tank **106** may not observe the actual performance of the laser **126** when targeted at other locations within the build tank **106**.

(14) In the example of FIG. 2, a beam measurement system **101** is depicted. The beam measurement system **101** can include a beam measurement device **104** positioned within the build tank **106**. The beam measurement device **104** can be used to inspect a beam profile of the laser **126** before or after parts **112** of FIG. 1 are built within the build tank **106**. In some embodiments, a control system **130** can be used to determine a placement position of the beam measurement device **104** within the build tank **106**. The control system **130** can be used to determine one or more commands for an adjustment of one or more aspects of the beam measurement device **104** to achieve alignment of a target with an intended measurement position. In embodiments, the control system **130** is configured to determine an intended measurement position to observe a laser beam profile of a laser **126** of the laser powder bed fusion system **100** of FIG. 1 and determine a target position of a beam measurement device **104** to align with the intended measurement position. In some embodiments, the control system **130** can determine positioning of other components, such as mirrors **105**. Operation of the laser **126** can be separately controlled.

(15) The laser **126** can be guided by one or more optical elements **140**, such as mirrors **105**, and be received as an incident beam **127** observed by the beam measurement device **104**. Various optic elements **140** external to the beam measurement device **104** can direct the incident beam **127** as a portion of the laser beam **126** into a port **142** of the beam measurement device **104**. The beam measurement device **104** can include an imager **144** and a plurality of optical elements **146** configured to form an optical path **148** to the imager **144** within the beam measurement device **104**. The optical elements **140**, **146** can include one or more mirrors and one or more beam dumps. Some of the optical elements **140**, **146** can be adjustable/movable while others can be fixed. As one example, the optical elements **146** within the beam measurement device **104** may be fixed, while the optical elements **140** external to the beam measurement device **104** can be movable. The number and position of the optical elements **140**, **146** depicted in FIG. 1 is for purposes of illustration and should not be construed as limiting. A path length of the optical path **148** within the beam measurement device **104** can be greater than a path P from the port **142** to a build plane **150**. The path P represents the path that the laser beam **126** would normally take to the build tank **106** if the beam measurement device **104** was not positioned within the build tank **106**. The path P may be equivalent to a height of the port **142** to the build plane **150**.

(16) Where adjustment of the optical elements **146** within the beam measurement device **104** is not possible, the position of the beam measurement device **104** can be adjusted by a positioning apparatus **151**. The positioning apparatus **151** can include, for example, an adapter plate **152** to mount the beam measurement device **104** to the positioning apparatus **151**. The positioning apparatus **151** can also include a goniometer **154** to set a tilt angle and/or measure a range of motion in tilting the beam measurement device **104**. The positioning apparatus **151** can also include a rotation stage **156** to rotate the beam measurement device **104** relative to the base plate **108**. The positioning apparatus **151** can also include an X-Y stage **158** to translate movement of the beam measurement device **104** relative to the base plate **108** in X-Y coordinates. The positioning apparatus **151** can be coupled to a mounting plate **160**. It will be understood that the positioning apparatus **151** can include other elements to support position adjustment and provide position feedback regarding the location and orientation of the beam measurement device **104**. Adjustments to the positioning apparatus **151** can be performed manually and/or automatically, for instance, as commanded by the control system **130**. In a manual mode, the command output of the control system **130** can be in terms of instructions or position settings to adjust the positioning apparatus **151** and/or one or more aspects of the beam measurement device **104** and/or optic elements **140**.

(17) The control system **130** can include a processing system **132** and a memory system **134**, where the memory system **134** stores executable instructions to configure the processing system **132** to perform a plurality of operations. The processing system **132** can include any type or combination of central processing unit (CPU), including one or more of: a microprocessor, a digital signal processor (DSP), a microcontroller, an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or the like. The memory system **134** can store data and instructions that are executed by the processing system **132**. In embodiments, the memory system **134** may include random access memory (RAM), read only memory (ROM), or other electronic, optical, magnetic, or any other computer readable medium onto which is stored data and algorithms in a non-transitory form. The control system **130** can also include an input/output (I/O) interface **136** operable to interface with various components of the laser powder bed fusion system **100** to sense, control, and communicate with components of FIG. **2** and other components, such as a user interface **138**. The user interface **138** can include a display, buttons, keys, switches, and other such interfaces for accepting user input and displaying output.

(18) FIG. **3** is a schematic view **200** of a build plate **108** of the laser powder bed fusion system **100** that includes multiple build locations **202** according to an embodiment. The build locations **202** can be offset in x-axis and y-axis directions relative to a surface of a base plate **108**. In the example of FIG. **3**, each of the build locations **202** can be defined in terms of X-Y-Z coordinates. A location **204** in X-Y-Z space may be computed by the control system **130** of FIG. **2** as a target position to align with the intended measurement position for each of the build locations **202**. A projection **206** between the build locations **202** and the location **204** can be used to establish one or more positioning adjustments and set one or more angles. Getting the length and angular positions set to match the laser beam profile at the build locations **202** can ensure that observations made by the imager **144** of FIG. **2** align with the laser beam profiles at each of the build locations **202**. The example of FIG. **3** is provided for purposes of explanation and is not limiting. A single part **112** of FIG. **1** can be manufactured at any of the build locations **202** or multiple parts **112** can be built in parallel at several of the build locations **202**. Multiple instances of the beam measurement device **104** can be used in parallel, for example, to support the use of multiple lasers **126** in parallel builds within the build tank **106** of FIG. **1**. In some embodiments, multiple lasers **126** can share a single instance of the beam measurement device **104**.

(19) FIG. **4** is a schematic view of an integrated beam measurement device **302** incorporated into a recoater **304** of a laser powder bed fusion system **300** according to an embodiment. The laser powder bed fusion system **300** can include an actuator **306** and a build plate **308**. The actuator **306** can control movement of the recoater **304** relative to the build plate **308**. For instance, the recoater **304** may be driven in a Y-axis direction, while the integrated beam measurement device **302** is driven to move in an X-axis direction, such that the integrated beam measurement device **302** can view any location on the build plate **308**. The control system **130**, as previously described with respect to FIG. **2**, can control movement of the recoater **304** and the integrated beam measurement device **302** relative to the build plate **308**. Further, the control system **130** can control one or more aspects within the integrated beam measurement device **302**. Although described in the example of FIG. **4** as being incorporated in recoater **304**, the integrated beam measurement device **302** can be incorporated into another movable component that can move in at least two axis directions (XY) relative to the surface above the build plate **308**.

(20) Observations made by the integrated beam measurement device **302** can be made on a layer-by-layer basis to provide inspection as the parts **112** of FIG. **1** are built. In some embodiments, multiple lasers **126** can share a single instance of the integrated beam measurement device **302**.

(21) FIG. **5** is a schematic view of an optical path **402** within the integrated beam measurement device **302** according to an embodiment. The integrated beam measurement device **302** can include one or more mirrors **404**, **406**, **408**, one or more beam dumps **410**, **412**, and an imager **414**. In the example of FIG. **5**, an incident beam **427**, similar to the incident beam **127** of FIG. **2**, can be

received from one or more lasers **126** (FIG. **1**). The mirror **404** can be a fixed, semi-transparent mirror that reflects a portion of the incident beam **427** along the optical path **402** to the mirror **406** and allows a remaining portion of the incident beam **427** to be absorbed by the beam dump **410**. The mirror **406** can be a moveable semi-transparent mirror that reflects a portion of the incident beam **427** along the optical path **402** to the mirror **408** and allows a remaining portion of the incident beam **427** to be absorbed by the beam dump **412**. As one example, a semi-transparent mirror can allow most of the laser energy of the incident beam **427** to be transmitted (e.g., not reflected) so the imager **414** does not get burned. The mirror **408** can be a fixed, fully reflecting mirror that directs the remaining portion of the incident beam **427** to the imager **414**.

(22) Accordingly, a series of mirrors **404**, **406**, **408** can be utilized within the integrated beam measurement device **302** to automatically adjust the path length of the optical path **402** within the integrated beam measurement device **302** to ensure that no matter where the target measurement location is, the laser beam profile is measured at the same path length that the laser would experience in normal operation (i.e., during the laser powder bed fusion printing process). At least one of these mirrors (e.g., mirror **406**) can be moveable to adjust the path length. The mirrors **404**, **406** can be semi-transparent mirrors such that the incident laser intensity is attenuated to appropriate levels before being imaged. Beam dumps **410**, **412** to dissipate the excess laser intensity may be included behind each semi-transparent mirror **404**, **406**. It will be understood that additional elements can be included in the integrated beam measurement device **302**. For instance, any number of optical elements can be included other than the specific configuration as depicted in the example of FIG. **5**.

(23) Turning to FIG. **6**, a flowchart shows a method **500** for additive manufacturing according to an embodiment. The method **500** of FIG. **6** is described with respect to FIGS. **1-6** and may include additional steps beyond those depicted in FIG. **6**. The method **500** can be controlled by a control system, such as the control system **130** of FIG. **2**. Some portions of the method **500** can be manually performed depending on the amount of automation available. The method **500** is described primarily in reference to the laser powder bed fusion system **100** of FIG. **1** and the laser powder bed fusion system **300** of FIG. **4** for purposes of explanation, although it will be understood that any of the components previously described can be used depending upon the system configuration.

(24) At block **502**, an intended measurement position to observe a laser beam profile of the laser powder bed fusion system **100**, **300** by a beam measurement device **104**, **302** can be determined, for instance, by the control system **130**. At block **504**, a target position of the beam measurement device **104**, **302** can be determined to align with the intended measurement position, for instance, by the control system **130**. At block **506**, an adjustment of one or more aspects of the beam measurement device **104**, **302** can be commanded to achieve alignment of the target with the intended measurement position. Commands can be automated or be in the form of instructions for positional settings for manual adjustments as needed. A positioning apparatus **151** can be configured to adjust a position of the beam measurement device **104** relative to a build plate **108** of the laser powder bed fusion system **100**. The actuator **306** is another example of a positioning apparatus that can be configured to adjust a position of the beam measurement device **302** relative to a build plate **308** of the laser powder bed fusion system **300**. At block **508**, the beam measurement device **104**, **302** can be operated to observe the laser beam profile after adjusting one or more aspects of the beam measurement device **104**, **302**.

(25) In some embodiments, the adjustment of one or more aspects of the beam measurement device **104** can include the positioning apparatus **151** changing an angle of the beam measurement device relative **104** to the build plate **108**. The angle can be defined in three-dimensional space. In some embodiments, the adjustment of one or more aspects of the beam measurement device **104**, **302** can include moving one or more of the optical elements, such as optical elements **140**, **146**, or mirrors **404**, **406**, **408**. The adjustment can maintain a path length of the optical path **148**, **402** within the

beam measurement device **104, 302** to accommodate a change in the target position. A path length of the optical path **148** within the beam measurement device **104** can be greater than a height of the port from a build plane **150** along path P. An optical path length within the beam measurement device can be adjustable to match an intended path length of a laser beam to the intended measurement position.

(26) In some embodiments, the laser beam profile can be analyzed at the target position as observed by the imager **144, 414**. The control system **130** can compare the laser beam profile to an expected beam profile and trigger an alert based on determining that the laser beam profile fails to match the expected beam profile within a comparison threshold. The alert can indicate that the part **112** need inspection or rework before completing the manufacturing process. Embodiments can support additive manufacturing for multiple parts **112** simultaneously.

(27) In summary, embodiments allow for the measurement of beam profile at any location across a build platform. This allows the user to ensure that the beam profile is within specification across the entire build platform. This knowledge will increase the certainty that the machine is operating as intended and may reduce the reliance on mechanical property specimens for certification of additive manufactured parts. Embodiments can also enable feedback control of the laser focus during the build. For multi-laser systems, beam profile measurement can ensure that the beam profile for different lasers acting at the same location are adequately similar, increasing consistency and uniformity of the process. This can also ensure that multiple beams are aligned to each other and the ability to correct inadequate alignment.

(28) A beam profile measurement device can be positioned such that it measures the laser beam in a state that it will be used during printing. To accomplish this, the beam profile measurement device is configured to work with the laser powder bed fusion machine of interest or be positioned using a secondary device tailored to the purpose. The positioning apparatus can place the measurement device at the appropriate angle and location such that the measurement occurs at the same path length as the laser would travel to impact the build plane. The total path length of the laser system, geometry of the laser powder bed fusion build chamber, optical path of the laser system, and target location on the build plane can be used to determine the appropriate location for the beam profile measurement device. It may be advantageous for the path length within the beam measurement device to be longer than the height of the device such that the device can be placed at a point in space above the build plane, allowing the device to measure out to the extents of the buildable area.

(29) In some embodiments, the laser powder bed fusion optics can utilize scanning galvanometers. The distance from the build plane to the first mirror and distance between mirrors can be computed/known. This information, coupled with the target X,Y,Z location of the measurement and the path length of the laser within the measurement device can allow for the calculation of the appropriate intermediate point and orientation in space at which the measurement device should be placed.

(30) The term “about” is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, “about” can include a range of $\pm 8\%$ or 5% , or 2% of a given value.

(31) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

(32) While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the

present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A laser beam measurement system comprising: a beam measurement device comprising an imager and a plurality of optical elements configured to form an optical path to the imager within the beam measurement device; a positioning apparatus configured to adjust a position of the beam measurement device relative to a build plate of a laser powder bed fusion system; and a control system configured to determine an intended measurement position to observe a laser beam profile of a laser of the laser powder bed fusion system, determine a target position of a beam measurement device to align with the intended measurement position, and command an adjustment of the positioning apparatus changing an angle of the beam measurement device relative to the build plate to achieve alignment of a target with the intended measurement position placing the beam measurement device at the angle and a location such that measurement occurs at a same path length as the laser would travel to impact a build plane.
2. The laser beam measurement system of claim 1, wherein an optical path length within the beam measurement device is adjustable to match an intended path length of a laser beam to the intended measurement position.
3. The laser beam measurement system of claim 1, wherein one or more of the optical elements are movable.
4. The laser beam measurement system of claim 3, wherein the adjustment maintains a path length of the optical path within the beam measurement device to accommodate a change in the target position.
5. The laser beam measurement system of claim 1, wherein the optical elements comprise one or more semi-transparent mirrors and one or more beam dumps.
6. The laser beam measurement system of claim 1, wherein a path length of the optical path within the beam measurement device is greater than a height of a port of the beam measurement device from the build plane.
7. The laser beam measurement system of claim 1, wherein the control system is configured to output the command to a user interface.
8. The laser beam measurement system of claim 1, wherein the control system is configured to analyze the laser beam profile at the target position as observed by the imager, compare the laser beam profile to an expected beam profile, and trigger an alert based on determining that the laser beam profile fails to match the expected beam profile within a comparison threshold.
9. A method for laser beam measurement in a laser powder bed fusion system, the method comprising: determining an intended measurement position to observe a laser beam profile of a laser of the laser powder bed fusion system by a beam measurement device; determining a target position of the beam measurement device to align with the intended measurement position; and commanding an adjustment of a positioning apparatus changing an angle of the beam measurement device relative to a build plate to achieve alignment of a target with the intended measurement position placing the beam measurement device at the angle and a location such that measurement occurs at a same path length as the laser would travel to impact a build plane, wherein the positioning apparatus is configured to adjust a position of the beam measurement device relative to the build plate of the laser powder bed fusion system.
10. The method of claim 9, wherein an optical path length within the beam measurement device is adjustable to match an intended path length of a laser beam to the intended measurement position.

11. The method of claim 9, wherein one or more of the optical elements are movable.
 12. The method of claim 11, wherein the adjustment maintains a path length of the optical path within the beam measurement device to accommodate a change in the target position.
 13. The method of claim 9, wherein the optical elements comprise one or more semi-transparent mirrors and one or more beam dumps, and at least one of the one or more semi-transparent mirrors is movable.
 14. The method of claim 9, wherein a path length of the optical path within the beam measurement device is greater than a height of a port of the beam measurement device from the build plane.
 15. The method of claim 9, further comprising: outputting the command to a user interface.
 16. The method of claim 9, further comprising: operating the beam measurement device to observe the laser beam profile after adjusting one or more aspects of the beam measurement device; analyzing the laser beam profile at the target position as observed by the imager; comparing the laser beam profile to an expected beam profile; and triggering an alert based on determining that the laser beam profile fails to match the expected beam profile within a comparison threshold.
 17. A laser beam measurement system comprising: a beam measurement device comprising an imager and a plurality of optical elements configured to form an optical path to the imager within the beam measurement device; a positioning apparatus configured to adjust a position of the beam measurement device relative to a build plate of a laser powder bed fusion system; and a control system configured to determine an intended measurement position to observe a laser beam profile of a laser of the laser powder bed fusion system, determine a target position of a beam measurement device to align with the intended measurement position, and command an adjustment of the beam measurement device relative to the build plate to achieve alignment of a target with the intended measurement position, wherein a path length of the optical path within the beam measurement device is greater than a height of a port of the beam measurement device from a build plane.
 18. The laser beam measurement system of claim 17, wherein an optical path length within the beam measurement device is adjustable to match an intended path length of a laser beam to the intended measurement position.
 19. The laser beam measurement system of claim 17, wherein the adjustment maintains a path length of the optical path within the beam measurement device to accommodate a change in the target position.
 20. The laser beam measurement system of claim 17, wherein the control system is configured to analyze the laser beam profile at the target position as observed by the imager, compare the laser beam profile to an expected beam profile, and trigger an alert based on determining that the laser beam profile fails to match the expected beam profile within a comparison threshold.
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